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265. What does the decoding of the instruction determine?

- a) The actions needed for execution
- b) The memory address of the instruction
- c) The result of the instruction
- d) The next instruction in the program

266. How are data needed for instruction processing fetched?

- a) From the Program Counter (PC)
- b) From the Instruction Register (IR)
- c) From the Control Unit Register (CUR)
- d) From a register or RAM through an address register

267. Where is the result of the instruction stored?

- a) Program Counter (PC)
- b) Instruction Register (IR)
- c) Address Register (AR)
- d) A register or a memory location

268. What triggers the continuation of the instruction cycle for the next instruction?

- a) Decoding of the current instruction
- b) Loading of the current instruction
- c) Incrementing the Program Counter (PC)
- d) Storing the result of the current instruction

269. When does the instruction cycle continue until there is no more instruction in the program?

- a) When the computer is turned off
- b) When an unrecoverable error occurs
- c) When the Program Counter (PC) reaches its maximum value
- d) When the last instruction is fetched and executed successfully

270. How many steps are involved in the instruction cycle while executing a program?

- a) Two steps
- b) Three steps
- c) Four steps
- d) Five steps

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Explanation

1. The correct answer is a) First generation. The first generation of computers, which emerged in the 1940s and 1950s, used vacuum tubes as their primary electronic component for circuitry.

2. The correct answer is a) Scientific calculations. The first generation of computers was primarily used for scientific and mathematical calculations, as well as for tasks such as code-breaking during World War II.

3. The correct answer is b) Second generation. The second generation of computers, developed in the late 1950s to the early 1960s, introduced transistors as a replacement for vacuum tubes. Transistors were smaller, more reliable, and generated less heat than vacuum tubes.

4. The correct answer is c) Third generation. The third generation of computers, which appeared in the mid-1960s, saw the introduction of integrated circuits (ICs). ICs combined multiple transistors and other electronic components on a single chip, resulting in smaller and more powerful computers.

5. The correct answer is c) Third generation. The third generation of computers saw the development of microprocessors, which are integrated circuits that contain the functions of a central processing unit (CPU) on a single chip. Microprocessors revolutionized computing by enabling the creation of smaller, faster, and more affordable computers.

6. The correct answer is c) Third generation. The development of high-level programming languages like COBOL (Common Business-Oriented Language) and FORTRAN (Formula Translation) occurred during the third generation of computers. These languages made it easier for programmers to write complex programs and contributed to the expansion of software applications.

7. The correct answer is d) Fourth generation. The use of graphical user interfaces (GUIs) and personal computers became prominent in the fourth generation of computers, which started in the late 1970s and continued into the present day. Personal computers with intuitive GUIs, such as the Apple Macintosh and the Microsoft Windows-based

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PCs, revolutionized computer usage by making them more accessible to non-technical users.

8. The correct answer is d) Fourth generation. The fourth generation of computers is characterized by the use of large-scale integration (LSI) and very large-scale integration (VLSI) circuits. LSI and VLSI refer to the integration of thousands or millions of transistors and other electronic components onto a single chip. This advancement in chip technology allowed for even greater computing power and miniaturization.

9. The correct answer is d) Fourth generation. The introduction of the internet and the World Wide Web (WWW) took place during the fourth generation of computers. The internet, as we know it today, emerged in the late 1960s and early 1970s, while the World Wide Web was invented in 1989 by Sir Tim Berners-Lee.

10. The correct answer is d) Fourth generation. Artificial intelligence (AI) and parallel processing became associated with the fourth generation of computers. During this period, advancements in hardware and software allowed for the development and application of AI technologies, such as expert systems and neural networks.

Parallel processing, which involves performing multiple computations simultaneously, also became feasible in the fourth generation, leading to significant performance improvements in various computing tasks.

11. The correct answer is c) Personal computer. A personal computer is designed to be used by a single user at a time. It is typically a desktop or laptop computer that provides computing capabilities and resources for individual users in homes, offices, or other personal settings.

12. The correct answer is b) Supercomputer. A supercomputer is specifically designed to perform complex calculations at incredibly high speeds. It is built with a large number of processors working in parallel to solve computationally intensive problems, such as weather forecasting, scientific simulations, or cryptography.

13. The correct answer is a) Mainframe computer. A mainframe computer is commonly used in large organizations for centralized data storage and processing. It is a powerful computer that can handle a vast amount of data and support multiple users simultaneously. Mainframes are known for their reliability, scalability, and ability to handle critical business applications.

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14. The correct answer is d) Embedded computer. An embedded computer is integrated into other devices or systems to perform specific tasks. It is designed to be a component of a larger system rather than a standalone computer. Embedded computers can be found in various devices such as smartphones, cars, appliances, and industrial equipment.

15. The correct answer is d) Laptop computer. A laptop computer, also known as a notebook computer, is characterized by its small size, portability, and battery power. Laptops are designed to be easily transported and used in different locations. They typically provide similar functionality to desktop computers but in a more compact form factor.

16. The correct answer is c) Personal computer. A personal computer is used for general-purpose computing tasks and is commonly found in homes and offices. It can be a desktop or laptop computer that provides computing capabilities and resources for individual users.

17. The correct answer is b) Supercomputer. A supercomputer is known for its high processing power, massive memory capacity,

and ability to handle complex simulations. Supercomputers are designed to solve computationally intensive problems that require substantial computational resources.

18. The correct answer is d) Embedded computer. An embedded computer is designed to perform specific tasks and is often used in industrial control systems. It is integrated into machines or equipment to control and monitor various processes and functions.

19. The correct answer is b) Supercomputer. Supercomputers are commonly used in scientific research, weather forecasting, and advanced calculations. Their high computational power and ability to process large datasets make them suitable for tasks that require extensive computational resources.

20. The correct answer is b) Supercomputer. Supercomputers are built with multiple processors and are used to solve complex problems by dividing the tasks among the processors. The parallel processing capabilities of supercomputers allow them to tackle highly demanding computational tasks efficiently.

21. The correct answer is a) Central Processing Unit (CPU). The CPU is responsible for executing

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instructions and performing calculations in a computer system. It is often referred to as the "brain" of the computer.

22. The correct answer is b) Keyboard. The keyboard is used to input data into a computer system by pressing keys corresponding to letters, numbers, and various other characters. It allows users to enter text, commands, and other input into the computer.

23. The correct answer is c) Hard Disk Drive (HDD). The hard disk drive is responsible for storing data and program instructions even when the computer is turned off. It provides non-volatile storage, meaning the data remains stored even without power.

24. The correct answer is a) Monitor. The monitor is responsible for displaying visual output from a computer system. It presents images, text, and other graphical information generated by the computer.

25. The correct answer is c) Hard Disk Drive (HDD). The hard disk drive is responsible for long-term storage of data and programs. It provides a large capacity for storing files and applications on a magnetic disk.

26. The correct answer is c) Printer. A printer is used to produce a permanent physical copy of information from a computer. It can print text, images, and other content onto paper or other media.

27. The correct answer is a) Power Supply Unit (PSU). The power supply unit is responsible for providing power to the computer system and regulating electrical flow. It converts the AC (alternating current) power from the wall outlet into the DC (direct current) power required by the computer's components.

28. The correct answer is d) Speakers. Speakers are responsible for producing sound output from a computer system. They allow users to hear audio, including music, voice, and system sounds.

29. The correct answer is a) Motherboard. The motherboard is responsible for managing and controlling the flow of data within the computer system. It provides the foundation for connecting various components, such as the CPU, RAM, storage devices, and peripherals.

30. The correct answer is c) Mouse. A mouse is used to control the movement of the cursor on a computer screen. It typically has buttons that can be clicked to select

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and interact with items displayed on the screen.

31. The correct answer is b) ROM (Read-Only Memory). ROM is a type of computer memory that retains its data even when the power is turned off. It contains pre-programmed instructions and data that are permanently stored during the manufacturing process.

32. The correct answer is a) Modem. A modem is responsible for converting digital signals to analog signals for transmission over telephone lines. It allows computers to communicate with each other over a telephone network by converting the digital data from the computer into a format suitable for transmission over analog telephone lines.

33. The correct answer is a) USB (Universal Serial Bus) port. USB ports are commonly used to connect external devices such as printers, scanners, keyboards, and mice to a computer. USB ports provide a standard interface for connecting and transferring data between the computer and external peripherals.

34. The correct answer is b) Motherboard. The motherboard is responsible for managing and coordinating the activities of all other hardware components in a

computer system. It provides the electrical connections and pathways for data transfer between the various components, including the CPU, RAM, storage devices, and expansion cards.

35. The correct answer is c) Optical Drive. An optical drive is used to read data from and write data to optical discs such as CDs and DVDs. It uses a laser to read and write data onto the disc surface, allowing users to access or store information on optical media.

36. The correct answer is c) Cache Memory. Cache memory is the fastest type of computer memory and is used to temporarily store data that the CPU needs to access quickly. It serves as a buffer between the CPU and main memory (RAM), providing faster access to frequently used instructions and data.

37. The correct answer is a) Power Supply Unit (PSU). The power supply unit is responsible for providing power to all other hardware components in a computer system. It converts the AC (alternating current) power from the wall outlet into the DC (direct current) power required by the computer's components.

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38. The correct answer is a) Monitor. A monitor is used to display text and graphics on a computer screen. It receives signals from the computer's graphics card and presents visual output to the user, allowing them to interact with the computer system.

39. The correct answer is a) USB (Universal Serial Bus) port. USB ports are commonly used to connect external devices such as keyboards, mice, printers, and other peripherals to a computer. They provide a versatile and widely supported interface for connecting a variety of devices to the computer.

40. The correct answer is b) Graphics Processing Unit (GPU). The GPU is responsible for processing and rendering images, videos, and animations in a computer system. It is designed to handle complex mathematical and graphical operations, making it well-suited for tasks related to graphics rendering and acceleration.

41. The correct answer is b) Program Counter (PC). The program counter holds the address of the next instruction to be executed in the CPU. It keeps track of the current position in the program and increments to the next instruction after each execution.

42. The correct answer is c) Arithmetic Logic Unit (ALU). The ALU is the part of the CPU that performs arithmetic and logical operations. It can handle tasks such as addition, subtraction, multiplication, division, and logical comparisons.

43. The correct answer is d) It controls and synchronizes the pace of operations. The control unit in the CPU is responsible for controlling and coordinating the activities of all the other components within the CPU and the overall operation of the computer. It ensures that instructions are fetched, decoded, and executed in the correct order and synchronizes the timing of operations.

44. The correct answer is a) Instruction decoding circuit. The instruction decoding circuit in the control unit interprets the instructions fetched from memory and determines the appropriate actions to be taken by other components in the CPU. It decodes the instruction's opcode and operands to execute the corresponding operation.

45. The correct answer is d) It improves computer performance by minimizing the need to access slow main memory. Cache memory is a small, fast memory located close to

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the CPU. Its purpose is to store frequently accessed data and instructions, reducing the time it takes to access them from the slower main memory. By keeping frequently used data closer to the CPU, cache memory helps improve overall system performance.

46. The correct answer is a) Number of clock cycles per second. The processing power of a CPU is commonly measured in terms of clock frequency, which represents the number of clock cycles executed by the CPU in a second. It is typically measured in hertz (Hz), megahertz (MHz), or gigahertz (GHz).

47. The correct answer is d) The time between two consecutive pulses emitted by the internal clock. A clock cycle in a computer system is the time interval between two consecutive pulses emitted by the internal clock of the CPU. It determines the basic timing unit for executing instructions and coordinating various operations within the CPU.

48. The correct answer is a) Hertz (Hz). The clock frequency of a computer system is measured in hertz (Hz), which represents the number of clock cycles per second. It indicates how many instructions and operations the CPU can execute in a given time frame.

49. The correct answer is d) A CPU with multiple processors on a single chip. A multi-core processor refers to a CPU that integrates multiple individual processor cores on a single chip. Each core can handle independent tasks, allowing for parallel processing and improved performance in multitasking environments.

50. The correct answer is a) Megahertz (MHz) or gigahertz (GHz). CPU speed is typically measured in terms of clock frequency and is expressed in megahertz (MHz) or gigahertz (GHz). It represents the number of clock cycles per second and provides an indication of the CPU's processing capabilities.

51. The correct answer is c) Random access memory (RAM). RAM is the component of the CPU that temporarily stores data values during processing. It provides fast access to data for the CPU to perform calculations and execute instructions. Unlike cache memory, RAM is not as fast but has larger capacity and retains its contents as long as power is supplied.

52. The correct answer is a) It holds the current instruction being executed. The instruction register (IR) in the CPU is responsible for holding the current instruction that is being executed. It stores the

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opcode (operation code) of the instruction and any associated operands, allowing the control unit to decode and execute the instruction.

53. The correct answer is c) Control and timing circuit. The control and timing circuit is the part of the CPU that produces control signals to direct all other parts of the computer. It coordinates the flow of data and instructions between different components of the CPU and ensures that operations are performed in the correct sequence and at the right time.

54. The correct answer is b) It performs arithmetic operations like addition and subtraction. The arithmetic logic unit (ALU) in the CPU is responsible for performing arithmetic and logical operations on data. It can handle tasks such as addition, subtraction, multiplication, division, logical comparisons, and bitwise operations.

55. The correct answer is a) Control unit (CU). The control unit in the CPU is responsible for controlling the transfer of data and instructions between main memory and registers. It coordinates and manages the execution of instructions, ensuring that the appropriate operations are performed and data is moved to the

appropriate locations within the CPU.

56. The correct answer is b) To coordinate the execution of instructions in the CPU. Clock cycles in a computer system are used to coordinate the execution of instructions in the CPU. The clock acts as a timing mechanism, synchronizing the activities of the CPU components and ensuring that instructions are processed at a regular rate.

57. The correct answer is a) Hertz (Hz). Hertz (Hz) is the basic unit used to measure the clock frequency of a computer system. It represents the number of clock cycles per second and indicates the speed at which the CPU can execute instructions and perform operations.

58. The correct answer is d) A CPU with multiple processors on a single chip. The term "multi-core processor" refers to a CPU that integrates multiple individual processor cores on a single chip. Each core is capable of executing instructions independently, allowing for parallel processing and improved performance in multitasking scenarios.

59. The correct answer is d) To improve computer performance by

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reducing memory access time.

Cache memory in a CPU is used to improve computer performance by reducing the time taken to access data from the slower main memory. It stores frequently accessed data and instructions closer to the CPU, providing faster access compared to main memory and reducing memory latency.

60. The correct answer is a) Number of clock cycles per second. The processing power of a CPU is typically measured in terms of clock frequency, which represents the number of clock cycles the CPU can execute in a second. It indicates the speed at which the CPU can perform operations and is commonly measured in hertz (Hz), megahertz (MHz), or gigahertz (GHz).

61. The correct answer is b) To store instructions and data temporarily. The purpose of a memory unit in a computer is to store instructions and data temporarily while the computer is in operation. It provides fast access to data for the CPU to perform calculations and execute instructions. The contents of the memory unit are volatile and are lost when power is turned off.

62. The correct answer is c) As sequences of bytes. Data is typically stored in computer memory as sequences of bytes. A byte is a unit of storage that consists of 8 bits.

Each byte can represent a character, a numeric value, or a portion of a larger data structure.

63. The correct answer is b) 8 bits. A single byte of memory consists of 8 bits. Each bit represents a binary digit (0 or 1). Bytes are the fundamental units of storage in computer memory, and they are used to represent and store data.

64. The correct answer is a) To store and manipulate data and instructions. Registers in the CPU are small, high-speed storage locations used to store and manipulate data and instructions during processing. They provide fast access to data for arithmetic and logical operations, and they hold the operands and results of calculations.

65. The correct answer is d) Random Access Memory (RAM) and Read-Only Memory (ROM). The two main types of memory used in a computer system are Random Access Memory (RAM) and Read-Only Memory (ROM). RAM is volatile memory used for temporary data storage during program execution, while ROM is non-volatile memory that stores permanent data and program instructions.

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66. The correct answer is a) Primary memory. Primary memory is also known as main memory. It refers to the memory directly accessible by the CPU and is used to store data and instructions during program execution. It includes RAM, cache memory, and other forms of high-speed memory.

67. The correct answer is a) It can be accessed directly and is volatile in nature. The primary characteristic of Random Access Memory (RAM) is that it can be accessed directly by the CPU. RAM is volatile memory, meaning that its contents are lost when power is turned off or interrupted. It allows for fast read and write operations and is used for temporary data storage.

68. The correct answer is b) Contents are destroyed. When power is switched off, the contents of RAM are destroyed. RAM is volatile memory, and it requires continuous power supply to retain its contents. When the power is cut off, the data stored in RAM is lost.

69. The correct answer is c) Cache memory. Cache memory is a type of memory that is limited in size, consumes low power, and is faster compared to secondary memory. It is located between the CPU and main memory and serves as a high-speed buffer for frequently

accessed data. Cache memory helps to reduce the average time required to access data and improve overall system performance.

70. The correct answer is c) A high-speed memory that stores frequently accessed data. Cache memory in a computer system is a high-speed memory that stores frequently accessed data. It is designed to provide fast access to data that is likely to be needed by the CPU in the near future. By storing frequently accessed data closer to the CPU, cache memory helps to reduce the average memory access time and improve system performance.

71. The correct answer is c) By allowing the processor to access data more rapidly than main memory. Cache memory improves the performance of a computer by allowing the processor to access data more rapidly than main memory. It stores frequently accessed data closer to the CPU, reducing the time needed to fetch data from slower main memory. This helps to minimize the CPU's idle time waiting for data and increases overall system efficiency.

72. The correct answer is b) To minimize the time required to search for data

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in the cache. The management goal of data stored in cache memory is to minimize the time required to search for data in the cache. This is achieved by employing efficient caching algorithms and data placement strategies that maximize the cache hit rate, which refers to the percentage of data accesses that result in a cache hit (finding the requested data in the cache).

73. The correct answer is not provided in the options. The fraction of time during which the data needed is found to be stored in cache memory is not specified in the given options. The cache hit rate refers to the percentage of data accesses that result in a cache hit. A higher cache hit rate indicates better cache performance and a lower need to access slower main memory.

74. The correct answer is c) Faster access time. The primary advantage of cache memory compared to main memory is faster access time. Cache memory is designed to provide quicker access to data compared to main memory. It stores frequently accessed data, allowing the CPU to retrieve it at a much faster rate. This helps to reduce the latency associated with fetching data from main memory.

75. The correct answer is c) By the number of instructions executed per second. The performance of a computer is typically measured by the number of instructions executed per second. This metric, known as instructions per second (IPS) or instructions per clock (IPC), reflects the speed and efficiency of the CPU in executing instructions. It indicates the computational capability of the computer system.

76. The correct answer is d) Address index. The term used for a sequence number that identifies a memory address is an address index. It represents the location or location identifier of data stored in memory. The address index allows the processor to access the desired data or instruction from memory by specifying the appropriate memory location.

77. The correct answer is a) Primary memory. Primary memory, also known as main memory, is accessed directly by the processor. It refers to the memory locations that can be directly addressed by the CPU. Primary memory includes RAM and other high-speed memory components that hold data and instructions during program execution.

78. The correct answer is b) Secondary memory. Secondary

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memory is expensive and limited in size compared to primary memory. Secondary memory refers to external storage devices such as hard disk drives (HDDs), solid-state drives (SSDs), and optical drives. These storage devices provide larger storage capacity but are slower compared to primary memory.

79. The correct answer is a) To provide a large storage area for data and programs. The purpose of secondary memory in a computer system is to provide a large storage area for data and programs. Unlike primary memory, which is volatile and limited in size, secondary memory offers non-volatile storage that persists even when the power is turned off. It allows for long-term storage of data and programs.

80. The correct answer is c) Cache miss rate. The term used to describe the fraction of time during which the cache memory is searched for data is the cache miss rate. It represents the percentage of data accesses that result in a cache miss (not finding the requested data in the cache) and require accessing slower main memory. A lower cache miss rate indicates better cache performance and reduced memory latency.

81. The correct answer is c) Large, non-volatile, and inexpensive storage. The main characteristic of

secondary memory is that it provides large storage capacity, is non-volatile (retains data even when power is turned off), and is relatively inexpensive compared to primary memory.

82. The correct answer is a) Read operation. The read operation retrieves data from memory. It involves accessing a specific memory location and retrieving the data stored at that location.

83. The correct answer is b) Write operation. The write operation stores data in memory. It involves writing or saving data to a specific memory location, replacing any existing data at that location.

84. The correct answer is b) Byte. The common unit of memory is the byte. A byte is a unit of storage that consists of 8 bits. It is used to represent a character, a numeric value, or a portion of a larger data structure.

85. The correct answer is b) 8 bits. There are 8 bits in a byte. Each bit represents a binary digit (0 or 1), and a byte consists of 8 of these bits.

86. The correct answer is b) Bytes. The maximum capacity of memory is typically measured in bytes. For example, you may have a computer

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system with a certain amount of memory specified in kilobytes (KB), megabytes (MB), gigabytes (GB), or even terabytes (TB).

87. The correct answer is a) 1024 bytes. There are 1024 bytes in a kilobyte (KB). This is based on the binary nature of computer systems, where 2^{10} (1024) is used as the approximation for kilo.

88. The correct answer is a) Terabyte (TB). The next unit of memory after a gigabyte (GB) is a terabyte (TB). A terabyte is equivalent to 1024 gigabytes.

89. The correct answer is a) Word size. The term "word size" refers to the number of bits a CPU can manipulate at one time. It determines the maximum size of data that can be processed in a single operation by the CPU.

90. The correct answer is b) 32 bits. The word size of a 32-bit processor is 32 bits. This means that the CPU can process 32 bits of data at a time.

91. The correct answer is b) It can process 32 bits of data at a time. A 32-bit processor means that the CPU can process 32 bits of data at a time. It refers to the size of the data bus and the registers in the CPU.

92. The correct answer is b) Faster access time in primary memory. Faster access time in primary memory contributes to increased computer performance. It reduces the time required for the CPU to fetch data and instructions from memory, resulting in faster overall processing.

93. The correct answer is c) To manipulate data during processing. The purpose of registers in the Arithmetic Logic Unit (ALU) is to manipulate data during processing. Registers hold operands and intermediate results of calculations, allowing the ALU to perform arithmetic and logical operations.

94. The correct answer is a) Secondary memory. Secondary memory is non-volatile in nature. It retains data even when the power is turned off. Examples of secondary memory include hard disk drives (HDDs) and solid-state drives (SSDs).

95. The correct answer is c) Copy operation. The copy operation allows for data copying. It involves reading data from a source location in memory and writing it to a destination location, creating a duplicate or copy of the data.

96. The correct answer is b) To logically access data from the memory. The purpose of the

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interface circuit in memory is to facilitate the logical

access of data from the memory. It provides the necessary communication and control signals between the memory and other components of the computer system.

97. The correct answer is a) Bit. The smallest block of memory is a bit, which represents a binary digit (0 or 1). Multiple bits are combined to form larger units of memory, such as bytes.

98. The correct answer is c) To store data for long periods of time. The purpose of secondary storage is to store data for long periods of time. It provides non-volatile storage that persists even when the power is turned off. Secondary storage devices, such as hard disk drives and solid-state drives, offer larger storage capacity than primary memory but with slower access times.

99. The correct answer is c) Kilobyte. The unit used to express the memory capacity of a computer system is typically the kilobyte (KB), along with larger units such as megabytes (MB), gigabytes (GB), and terabytes (TB). These units represent increasing powers of

1024, reflecting the binary nature of computer systems.

100. The correct answer is a) To store frequently accessed data. The purpose of cache memory is to store frequently accessed data. It acts as a high-speed buffer between the CPU and main memory, allowing the CPU to quickly access frequently used data without having to retrieve it from slower main memory. Cache memory helps improve overall system performance by reducing memory latency and increasing the speed of data access.

101. The correct answer is c) Primary memory. Primary memory, also known as main memory or RAM (Random Access Memory), is directly accessed by the processor during program execution. It stores the instructions and data that the processor needs to perform its tasks.

102. The correct answer is c) Non-volatile storage and low cost. The main characteristic of secondary memory is that it provides non-volatile storage, meaning it retains data even when the power is turned off. Secondary memory, such as hard disk drives and solid-state drives, generally offers larger storage capacity than primary memory but with slower access times. It is also typically less

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expensive per unit of storage compared to primary memory.

103. The correct answer is b) Read. The read operation retrieves data from memory. It involves accessing a specific memory location and retrieving the data stored at that location.

104. The correct answer is a) Bit. The smallest unit of memory is a bit, which represents a binary digit (0 or 1). Multiple bits are combined to form larger units of memory, such as bytes.

105. The correct answer is c) To speed up data access by storing frequently used data. The purpose of cache memory in a computer system is to speed up data access. It works by storing frequently used data closer to the processor, allowing for faster retrieval compared to accessing the data from main memory. Cache memory acts as a buffer between the CPU and main memory, reducing memory latency and improving overall system performance.

106. The correct answer is b) Gigabytes (GB). The memory capacity of a computer system is typically expressed in units such as kilobytes (KB), megabytes (MB), gigabytes (GB), and larger units like terabytes (TB) or petabytes (PB).

These units represent increasing powers of 1024, reflecting the binary nature of computer memory.

107. The correct answer is a) Primary memory. Primary memory, or RAM, is volatile in nature. It requires a continuous power supply to retain data. When the power is turned off, the data stored in primary memory is lost.

108. The correct answer is d) To logically access data from the memory. The interface circuit in memory facilitates the logical access of data from the memory. It provides the necessary communication and control signals between the memory and other components of the computer system, allowing the CPU to read from and write to specific memory locations.

109. The correct answer is b) Byte. The unit commonly used to express the memory capacity of a computer system is the byte. A byte consists of 8 bits and is used to represent a character, a numeric value, or a portion of a larger data structure.

110. The correct answer is d) 64 bits. The word size of a 64-bit processor is 64 bits. This means that the CPU can process 64 bits of data at a time, allowing for larger data manipulation and addressing

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capabilities compared to processors with smaller word sizes.

111. The correct answer is c) It consists of primary memory, cache memory, and secondary memory. The memory hierarchy in a computer system typically consists of multiple levels of memory. Primary memory (such as RAM) is the fastest and closest to the CPU, cache memory is the next level and acts as a buffer between the CPU and main memory, and secondary memory (such as hard disk drives) is the largest and slowest level of memory. Virtual memory, represented by option d, is a technique that uses a combination of primary memory and secondary storage to provide the illusion of a larger memory capacity.

112. The correct answer is b) Cache memory. Among the levels of the memory hierarchy, cache memory has the fastest access time. It is designed to provide rapid access to frequently used data by storing a copy of recently accessed data from primary memory.

113. The correct answer is c) To improve the performance of memory access. The main purpose of the memory hierarchy in a computer system is to improve the overall performance of memory access. By utilizing different levels of memory with varying access times

and capacities, the memory hierarchy aims to provide faster and more efficient access to data for the CPU, thereby enhancing the system's performance.

114. The correct answer is b) Cache memory. Cache memory is the type of memory closest to the CPU in the memory hierarchy. It is designed to be located on the same chip as the CPU or in close proximity, allowing for faster data access compared to primary memory or secondary memory.

115. The correct answer is c) Small capacity and fast access time. Cache memory typically has a smaller capacity compared to primary memory or secondary memory. However, it compensates for the smaller capacity by offering faster access times. The purpose of cache memory is to store frequently accessed data and instructions to reduce the time needed to access them from slower levels of memory.

116. The correct answer is c) Secondary memory. Secondary memory, such as solid state drives (SSDs) or hard disk drives (HDDs), is typically used at the level of secondary memory in the memory hierarchy. Secondary memory provides a large storage capacity at a slower access time compared to

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primary memory and cache memory.

117. The correct answer is a) To provide additional storage capacity beyond the physical memory. The role of virtual memory in the memory hierarchy is to provide additional storage capacity beyond the physical memory (RAM). It achieves this by utilizing a portion of the secondary storage (such as the hard disk) as an extension of the physical memory. Virtual memory allows the system to run programs that require more memory than is physically available.

118. The correct answer is c) Secondary memory. Among the levels of the memory hierarchy, secondary memory has the largest storage capacity. Secondary memory, such as hard disk drives, can provide storage in the order of terabytes or even petabytes, allowing for the long-term storage of large amounts of data.

119. The correct answer is a) Primary memory. Primary memory, such as RAM, typically has the highest cost per byte of storage compared to cache memory, secondary memory, or virtual memory. Primary memory is built using faster and more expensive technologies, which contribute to its higher cost.

120. The correct answer is d) Through a controlled movement based on data usage patterns. Data is transferred between different levels of the memory hierarchy through a controlled movement based on data usage patterns. This process involves algorithms and strategies to determine which data should be moved to higher levels of the hierarchy (such as cache memory) for faster access and which data can be moved to lower levels or secondary storage when it is less frequently accessed. The movement of data is managed by hardware and software components responsible for memory management within the computer system.

131. The correct answer is d) Expansion bus. The expansion bus is responsible for connecting expansion cards (such as graphics cards, sound cards, network cards) to the motherboard. Expansion cards provide additional functionality to the computer system and connect to the expansion slots on the motherboard.

132. The correct answer is d) Provide a way to connect additional components to the motherboard. The primary purpose of the expansion bus is to provide a means of connecting additional

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components, such as expansion cards or peripherals, to the motherboard. It allows for the expansion and customization of the computer system by adding new functionality or features.

133. The correct answer is b) Power supply unit. The power supply unit (PSU) is responsible for managing the cooling system of the computer. It supplies power to the various components of the computer system, including fans and cooling mechanisms, to ensure proper cooling and prevent overheating.

134. The correct answer is b) Address bus. The address bus carries memory addresses, which are used to identify specific locations in the memory. The CPU uses the address bus to specify the location of data or instructions it wants to read from or write to in the main memory.

135. The correct answer is a) Report the status of various devices. The control signals carried by the control bus are used to report the status of various devices in the computer system. These signals allow for communication between different components and facilitate control and coordination of operations.

136. The correct answer is a) Through the device controllers. Devices are connected to the computer's internal bus through device controllers. Device controllers act as intermediaries between the devices and the bus, managing the communication and data transfer between the devices and the computer system.

137. The correct answer is c) Provide power to all components of the computer. The power supply unit (PSU) in a computer system is responsible for providing power to all components. It converts the incoming AC power from the electrical outlet into the appropriate DC power required by the computer's internal components.

138. The correct answer is c) Random access memory (RAM). RAM is the component on the motherboard that is responsible for temporarily storing data and instructions during processing. It provides fast and temporary storage that the CPU can access quickly, allowing for efficient execution of programs.

139. The correct answer is c) It allows easy connection and integration of components. The advantage of using a motherboard in a computer system is that it allows for easy connection and

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integration of components. The motherboard provides physical connectors and buses that enable the various components, such as the CPU, memory, storage devices, and expansion cards, to communicate with each other and work together as a cohesive system.

140. The correct answer is d) By providing physical connections and buses. The motherboard facilitates communication between the CPU and memory by providing physical connections and buses. The CPU is directly connected to the motherboard, and the motherboard has the necessary buses, such as the address bus and data bus, to enable data transfer between the CPU and memory modules. The motherboard acts as a central hub for communication and coordination between the CPU, memory, and other components.

141. The correct answer is c) Connect all hardware components. The motherboard serves as the main circuit board in a computer system and provides physical and electrical connections for all the hardware components, such as the CPU, RAM, storage devices, expansion cards, and peripherals.

142. The correct answer is c) Power supply unit (PSU)
The power supply unit (PSU) is a separate component that provides

power to the motherboard and other hardware components. It is usually not integrated into the motherboard itself.

143. The correct answer is d) CPU sockets
The CPU is connected to the motherboard using CPU sockets, which provide the physical and electrical interface between the CPU and the motherboard. Different CPU sockets are designed to support specific CPU models.

144. The correct answer is c) Integrated circuits (ICs)
Integrated circuits (ICs) are electronic components that contain multiple interconnected transistors, resistors, and capacitors on a single chip. These ICs form the main circuitry on the motherboard and perform various functions such as data processing and control.

145. The correct answer is d) Expansion bus
The expansion bus is a bus system on the motherboard that allows expansion cards (such as graphics cards, sound cards, and network cards) to be connected and communicate with the CPU and other components.

146. The correct answer is c) Initialize the hardware and start the operating system